HUMIDITY SENSORS READING PREDICTION AND INCUBATOR PARAMETER DETERMINATION

INTERNALLY FUNDED STUDENT PROJECT REPORT - YEAR

SUBMITTED BY

BIANCAA.R

II year

3122223002025

Supervisor

Dr.Partibane .B

BE, M.Tech,Phd.

Associate professor, ECE Department

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
SSN COLLEGE OF ENGINEERING
KALAVAKKAM-603110

Date: 5.8.24

CERTIFICATE OF COMPLETION

This is to certify that the project titled **HUMIDITY SENSORS READING PREDICTION AND INCUBATOR PARAMETER DETERMINATION** undertaken by **Biancaa.R** has been completed as per the proposed aim and objectives.

Faculty In-charge	**Head, ECE Department**
Dr.B Partibane	Dr. P. Vijayalakshmi
Associate professor	Professor

Date:

CERTIFICATE OF UTILISATION

This is to certify that an amount of Rs. 20000/- (Rupees only) was sanctioned for the project titled **HUMIDITY SENSORS READING PREDICTION AND INCUBATOR PARAMETER DETERMINATION** and an amount of Rs. Twenty thousand was utilised for the purpose for which it was sanctioned.

Team Members:

1. Biancaa.R 3122223002025

Faculty In-charge

Dr.B. Partibane
M.Tech.(Ph.D)
Associate Professor, ECE

Head, ECE Department

Dr. P Vijayalakshmi

Professor

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Proof of sanction:

Sanctioned Internally Funded Projects for First Year UG/PG Students 2023

SRI SIVASUBRAMANIYA NADAR COLLEGE OF ENGINEERING KALAVAKKAM-603110

ECE - UG STUDENT PROJECTS

S. No	Name and Year of the Student(s)	Project Guide(s)	Title of the Project	Duration	Budget & Items Approved
1	Aathi Shankar K.S. Jayanth J. Sanjay Kumar	Dr. M. Srinivasan SSN RC	Enhancing the thermoelectric properties if mineral acanthite (Ag2S) by filler compounds for wearable thermoelectric generators	2 Years	**Budget: Rs.25000** **Items Approved:** 1. Chemicals 2. Characterization 3. Miscellaneous
2	I. Gowshika R. Prithika	Dr. P. Rajesh Physics	Cobalt-Manganese-Zinc Oxide on silicon carbide composite electrode for fabrication of symmetric supercapattery	1 Year	**Budget: Rs.20000** **Items Approved:** 1. Chemicals 2. Miscellaneous
3	R. Biancaa	Dr. S. Kirubaveni	Analytical approach of the effect of surfactant on mixed metal oxide materials (ZnO/NiO) to enhance the sensitivity of a humidity sensor	1 Year	**Budget: Rs.20000** **Items Approved:** 1. BET Studies 2. FESEM and XPS Studies 3. Chemicals 4. Miscellaneous

PROJECT OVERVIEW

1. PROJECT TITLE

   HUMIDITY SENSORS READING PREDICTION AND INCUBATOR PARAMETER DETERMINATION

2. PROJECT MEMBERS

   Biancaa.R

3. PROJECT GUIDE

   Dr.B. Partibane
   M.Tech.,Ph.D
   Associate Professor

   Dept. of ECE

4. MAJOR RESEARCH AREA

5. PROJECT DURATION

Date of Sanction of Project:	13.3.23
Date of Submission of Project:	15.3.24
Project Duration:	1 year

6. PROJECT BUDGET

   Project funded by SSN Trust

Fund Sanctioned:	Rs. 20000
Fund Used:	Rs. 20000

CHAPTERS

Humidity sensor reading prediction and incubator parameter determination

ABSTRACT:

This paper presents a proposed system designed for the development of a humidity sensor characterized by enhanced accuracy and reduced response time when compared to commercially available humidity sensors. The primary objective of the system is to ascertain the humidity within an incubator, wherein a preterm infant is housed. The core components of the system include the fabricated humidity sensor, a temperature sensor, and an incubator, collectively employed for continuous monitoring of pertinent parameters.

The research endeavours to employ machine learning algorithms to prognosticate optimal settings for incubator functionality, with an emphasis on achieving a high success rate in regulating temperature and humidity. Furthermore, the study extends to predicting the average values of breath rate and breath volume of the infant. Health factors such as the infant's weight in grams and gestational age are also considered in the predictive model.

The final predictive model developed and deployed had used linear regression which outperformed its counterparts ridge regression and decision tree models with a mean absolute error of 1.24.The experimental setup involves integrating the fabricated humidity sensor and a commercially available temperature sensor into an Arduino board to facilitate the determination of crucial parameters. This comprehensive approach seeks to enhance the efficiency of incubator systems for preterm infants through the integration of advanced sensing technologies and predictive modelling.

1.Introduction:

Preterm birth, commonly referred to as premature birth, denotes the delivery of an infant prior to reaching 38 weeks of gestational age, accompanied by a birth weight below two kilograms. This phenomenon signifies that the neonate has undergone an abbreviated period of intrauterine development and, consequently, is not fully equipped for independent survival. The truncated gestational period increases the likelihood of complex medical challenges, necessitating specialized care. Premature infants are particularly susceptible to a spectrum of health complications, with respiratory-related ailments ,pulmonary ailments being predominant. The underdeveloped nature of their lungs renders them more vulnerable to the exigencies of the external environment. This vulnerability underscores the critical need for meticulous monitoring and specialized interventions to address the unique medical exigencies associated with preterm birth.

In the year 2021, the incidence of preterm birth affected approximately one in every ten infants born in the United States, highlighting the prevalence and societal impact of this issue (source: Centers for Disease Control and Prevention, https://www.cdc.gov/reproductivehealth/maternalinfanthealth/pretermbirth). It is imperative to recognize that the prevalence of premature births surpasses general awareness, emphasizing the imperative for continued research, public health initiatives, and medical interventions to ameliorate the outcomes and well-being of preterm infants and their families.

2.Literature review:

There are various new methodologies for designing incubators recently. For example Dive and Kulkarni designed an incubator that can detect the light luminance inside the incubator and the audio voice of the premature baby in the incubator[1].The proposed incubator system can notify the doctor and nurse about the baby's condition. When the baby cries for any reason, the alarm is turned on and will go on until it is manually turned off.

In a study by Atul Kale ,They had used GSM technology to send alerts about the baby's condition.[2] . The Arduino board, in conjunction with a Sim 800 GSM Module as an information transmission submodule, delivers SMS to a medical expert the baby body's sensor data monitored by the system. In summary, this study provided new ways for designing a dependable baby hatchery using the UNO microcontroller.

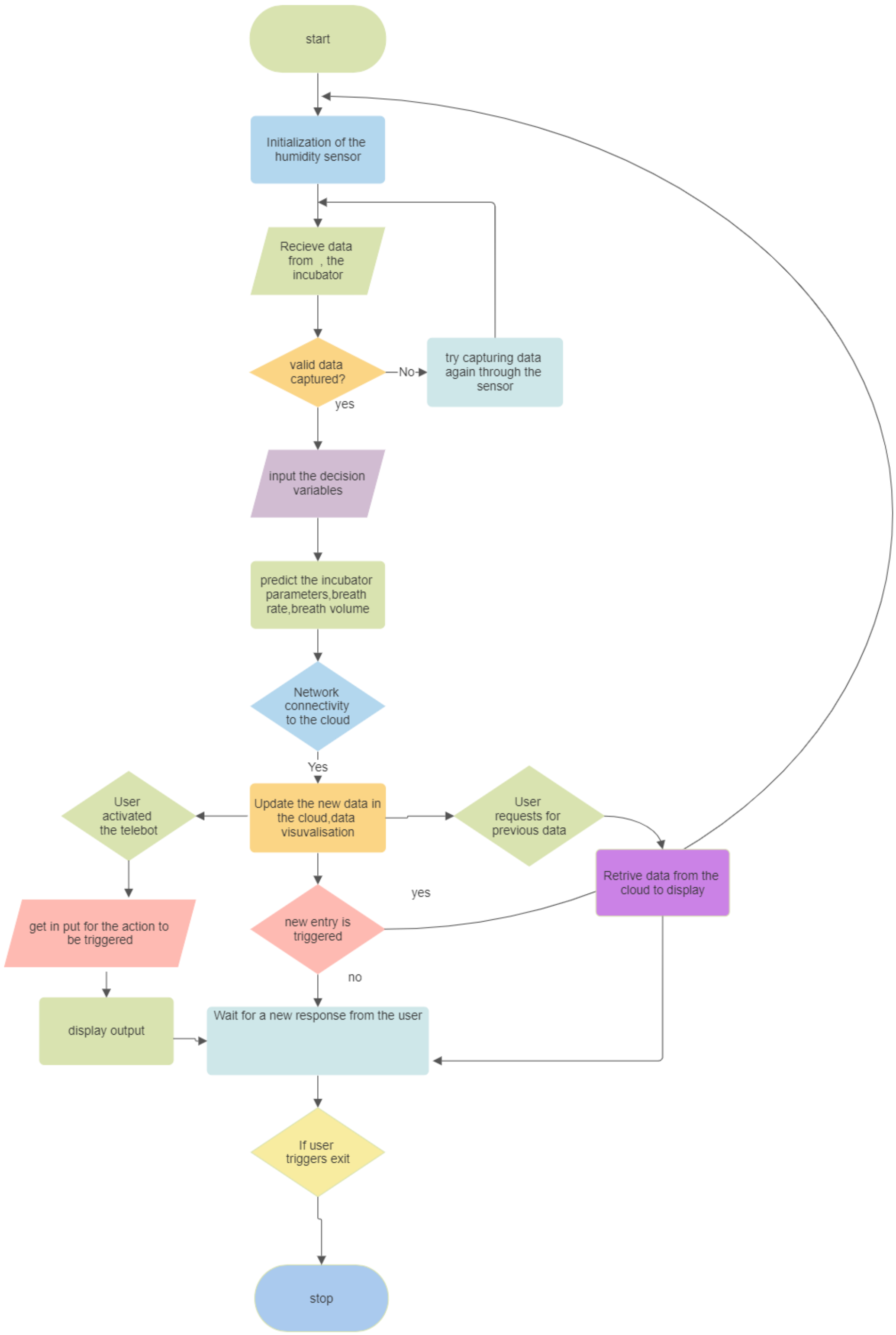
In the paper An android application based temperature and humidity monitoring and controlling child incubators paper by Mohaiminul Islam and Navila Rahman, preexisting dht11 had been used along with hc-06 and an interconnection of 2 incubators had been done wirelessly.(by means of the Bluetooth based android application)

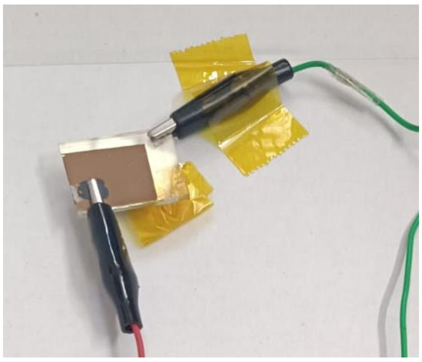
Non-contact physiological monitoring of preterm infants in the Neonatal Intensive Care Unit Mauricio Villarroel1*, Sitthichok Chaichulee 1 , João Jorge 1 , Sara Davis2 , Gabrielle Green2 , Carlos Arteta3 , Andrew Zisserman 3 , Kenny McCormick2 , Peter Watkinson4 and Lionel Tarassenko1 In this paper no sensors were used for the determination of parameters rather a contemprery approach of camera image capturing and use of image analysis models like cnn,naïve bayes were used to determine the parameters(multi-task Convolutional Neural Network (CNN), time periods during which the infant was present or absent from the incubator were automatically detected from the video recordings. Regions of interest (ROI) corresponding to skin were segmented from each video frame. These ROIs were used to extract cardiac-synchronous Photoplethysmographic Imaging (PPGi) and respiratory signals, from which heart rate and respiratory rate were estimated.)[11]

[12]Multiple Fault Detection and Smart Monitoring System Based on Machine Learning Classifiers for Infant Incubators Using Raspberry Pi 4(using a microprocessor) Maryam A. Mahdi*, Sabah A. Gittaffa, Abbas H. Issa - The proposed system had used four temperature sensors that were later extended to monitor skin temperature and two humidity sensors,The infant was monitored.

[13] Shabeeb et al. [12] proposed a method by which the humidity levels and air temperature in the infant's incubator are remotely monitored as a result of using an Arduino microcontroller with various sensors (DHT11/DHT22) and Internet of Things (IoT) applications. Using a wireless (ESP8266Wi-Fi) connection, the system connects to a network and is linked to a smartphone or computer application.

3.Workflow diagram:





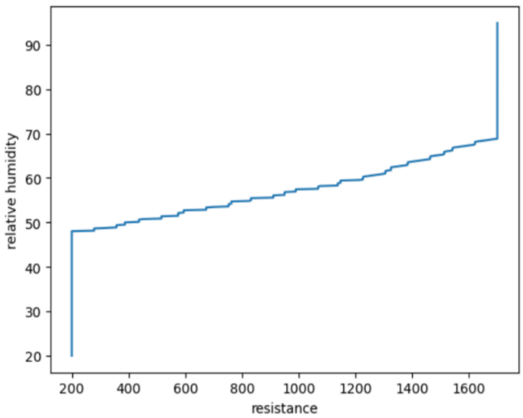
3.Components used :

3.1 Fabricated humidity sensor:

Within the structure of the fabricated humidity sensor, the positive segment is comprised of PEDOT:PSS, serving as the p-type material of the electrical connection. Conversely, the glass substrate is coated with a mixture of ZnO and NiO, functioning as the N-type material and serving as the negative terminal of the connection.

The escalation in humidity levels is denoted by an augmentation in oxygen ions, consequently resulting in heightened current conduction. Metal contacts are affixed to both terminals, thereby establishing a conductive pathway for electrical signals. This intricate arrangement facilitates the sensor's capacity to discern variations in humidity levels through the modulation of current conduction, offering a nuanced and responsive mechanism for humidity detection.

Fig 3.1.1 fabricated humidity sensor



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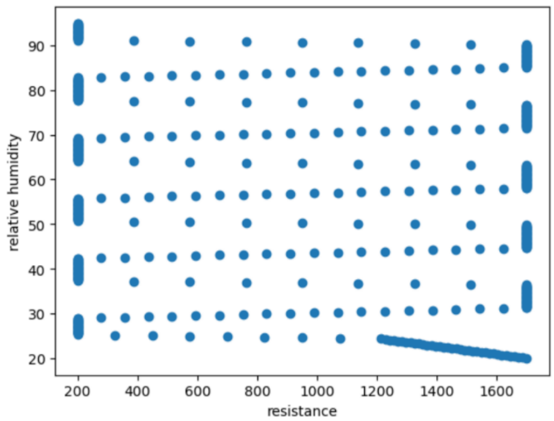


Fig 3.1.2: Graphical relation between resistance, relative humidity

By employing the aforementioned rationale, it becomes plausible to establish a quantitative relationship between the measured resistance in the humidity sensor and the corresponding relative humidity. The constructed models encompassed linear regression, ridge regression, and decision tree methodologies. Following a rigorous assessment, testing the linear regression model exhibited superior performance, consequently being selected as the definitive model for deriving an expression correlating the resistance measurement with the ambient relative humidity. This selected linear regression model was subsequently utilized for the conclusive testing phase, affirming its efficacy in providing a robust and accurate representation of the intricate relationship between sensor resistance and ambient relative humidity levels of the surroundings. The use of machine learning techniques is justified as predicting an setting of optimum values of temperature and humidity by human medical practitioners may sometimes lead to faulty and unfavourable outcomes. Due to the training of the model with a variety of cases and their outcomes the setting of these parameters would be far more suitable and tailormade for the infant inhabiting the incubator system. In the dataset a 20/80 ratio was used for test train split for checking the performance and accuracy of the developed model.

Confusion matrix for the various models built:

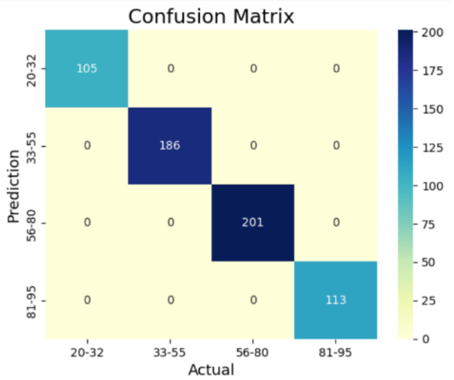


Fig 3.1.3 Confusion matrix linear regression

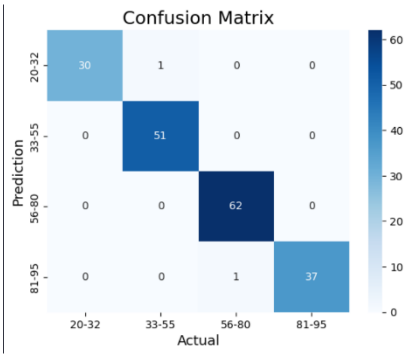


Fig 3.1.3 Cm decision tree

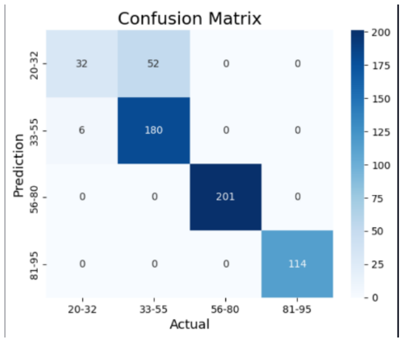


Fig 3.1.4 Confusion matrix ridge

Evaluation of the models built for the various devices:

Regression Model	Device Name														
	SD:CT-3					SD:CT-4					SD:CT-5				
	MSE	RMSE	MAE	R^2	Acc.	MSE	RMSE	MAE	R^2	Acc.	MSE	RMSE	MAE	R^2	Acc.
Decision Tree	15.5	3.93	0.41	0.99	0.98	15.5	3.93	0.41	0.99	0.98	15.5	3.93	0.41	0.99	0.98
Ridge Reg.	19.8	4.4	1.5	0.95	0.87	4.22	2.05	0.5	0.99	0.88	8.72	2.95	0.92	0.98	0.89
Linear Reg.	$2.3e^{-24}$	$1.52e^{-12}$	1.24	0.99	1.0	$2.1e^{-24}$	$1.4e^{-12}$	$1.7e^{-12}$	0.99	0.99	$2.4e^{-24}$	$1.5e^{-19}$	$1.3e^{-12}$	0.99	1.0

Fig 3.1.2 The model evaluation for the various devices fabricated

1.	precision	recall	f1-score	support
1	1.00	1.00	1.00	105
2	1.00	1.00	1.00	186
3	0.97	0.99	1.00	201
4	1.00	1.00	1.00	113
2. accuracy			0.98	605
3. macro avg	0.98	0.99	1.00	605
4. weighted avg	1.00	1.00	1.00	605

Chart3.1.1 Classification report of linear regression model

1.	precision	recall	f1-score	support
i. 1	0.88	0.51	0.65	84
ii. 2	0.81	0.97	0.88	186
iii. 3	1.00	1.00	1.00	201
iv. 4	1.00	1.00	1.00	114
2. accuracy			0.92	585
3. macro avg	0.92	0.87	0.88	585
4. weighted avg	0.92	0.92	0.91	585

Chart3.1.2 Classification report of ridge regression model

i.	precision	recall	f1-score	support
b. 20-32	0.97	1.00	0.98	30
c. 32-55	1.00	0.98	0.99	52
d. 55-80	1.00	0.98	0.99	63
e. 80-95	0.97	1.00	0.99	37
2. accuracy			0.99	182
3. macro avg	0.99	0.99	0.99	182
4. weighted avg	0.99	0.99	0.99	182

Chart3.1.3 Classification report of decision tree model

The various parameters used to evaluate the model is as follows:

1. Accuracy

- **Definition:** Measure of the ratio of correct predictions to the total number of forecasts in model performance. A higher accuracy indicates a lower error rate.
- **Formula:** Accuracy = (TP + TN) / (TP + TN + FP + FN)

2. Precision

- **Definition:** The proportion of true positive predictions relative to the sum of true positive and false-positive predictions. Indicates how many positive predictions were correct.
- **Formula:** Precision = TP / (TP + FP)

3. Recall

- **Definition:** The ratio of true positive predictions to the sum of true positive and false-negative predictions. Represents how many true positives were predicted or how many correct hits were found.
- **Formula:** Recall = TP / (TP + FN)

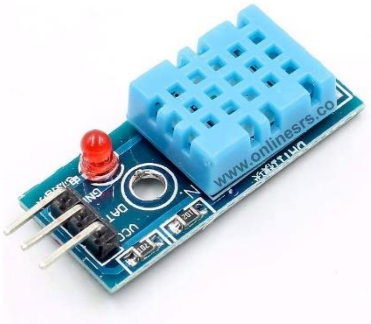
4. F1-Score

- **Definition:** A statistical metric for evaluating model performance using the harmonic mean of precision and recall.
- **Formula:** F1 = (2 × Precision × Recall) / (Precision + Recall)

Where:

- **TP (True Positive):** Predicted value is positive, and the actual one is positive.
- **TN (True Negative):** Predicted value is negative, and the actual one is negative.
- **FP (False Positive):** Predicted value is positive, but the actual one is negative.
- **FN (False Negative):** Predicted value is negative, but the actual one is positive.

These metrics collectively provide a comprehensive assessment of the model's performance, covering accuracy, precision, recall, and the F1-score.



3.2 DHT11 Humidity sensor (humidity reference): The DHT11 sensor, characterized by its economical nature, makes it a versatile and cost effective solution. It serves the purpose of assessing both humidity and temperature values. Within the experimental configuration, it was employed as a benchmark reference to validate the humidity measurements obtained from the fabricated humidity sensor and concurrently ascertain the internal temperature within the incubator.

Fig 3.2.1 dht11

The DHT11 sensor utilizes a capacitive humidity sensor and a thermistor to gauge the ambient air's humidity and temperature, respectively.

The capacitive humidity sensor is responsible for determining the humidity level by measuring the changes in capacitance resulting from the absorption or release of water vapor by a hygroscopic material.

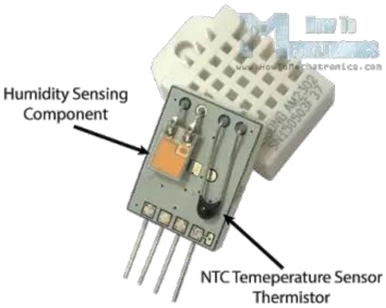


Fig 3.2.2 Internal structure of dht11

Concurrently, a thermistor, a temperature-sensitive resistor, is employed to measure the temperature of the surroundings. The thermistor's resistance changes with variations in temperature, and this alteration is then utilized to derive the corresponding temperature value.

These internal components are integrated within the sensor's housing, creating a compact and efficient unit.

3.3 Arduino UNO Board:

Arduino is an open source computer hardware and software company single-board microcontrollers and microcontroller kits for building digital devices ,with all the various functionalities in built. Even when a microcontroller was not as capable as a microprocessor say for instance raspberry pi 4, The basic functionality and the control over devices was sufficient enough to make this project feasible,and interactive objects that can sense and control objects in the physical and digital world. Arduino board designs use a variety of microprocessors and controllers. The boards are equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards or Breadboards (shields) and other circuits. The boards feature serial communications interfaces, including Universal Serial Bus (USB) on some models, which are also used for loading programs from personal computers. The Arduino Uno is a microcontroller board based on the ATmega328. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started.

4.Results and discussion:

The proposed infant incubator system:

In the proposed infant incubator system, the primary objective is to forecast and regulate crucial incubator parameters, specifically the temperature and relative humidity, to optimize the conditions for the well-being of the infant housed within the incubator. This predictive framework aims to determine the optimal setpoints for temperature and humidity, thereby maximizing the likelihood of favorable outcomes for the infant and to increase the chances of survival.

This conceptual framework can be extrapolated to hatcheries, broadening its applicability. By leveraging the predicted parameter values, the system ensures that the infant, or in the case of hatcheries, the developing organisms, inhabit an environment tailored for optimal growth and development. This predictive approach underscores a proactive strategy to maintain an ideal incubator environment, contributing to enhanced efficacy and success rates in the care of infants and hatchery operations

4.1The prediction of the incubator parameters are based on:

4.1.1 Age of the Infant: The age of the infant, quantified in gestational weeks, serves as a pivotal determinant of the infant's health status. A positive correlation exists between gestational age and the overall well-being of the infant, with a greater gestational age contributing to heightened health and increased chances of survival.

4.1.2 Weight of the Infant: The weight of the infant, measured in grams, functions as an approximate indicator of the infant's general growth and developmental status. This metric is instrumental in estimating the requisite parameters for the incubator, facilitating a nuanced approach to providing optimal conditions based on the infant's weight.

4.1.3 Room Temperature: The anticipated values of incubator parameters are subject to variation based on the ambient room temperature. Acknowledging and incorporating this external factor is essential for the accurate prediction and regulation of incubator conditions.

4.1.4 Room Humidity: Similar to room temperature, the projected values of incubator parameters are contingent upon the surrounding room humidity. Recognizing and accounting for this environmental variable is critical for refining the precision of the predicted incubator conditions.

4.1.5 Heat Out of Incubator: The heat dissipated during the operation of the incubator constitutes the heat outflow from the system. This factor is integral to comprehending the overall thermal dynamics of the incubator, thereby contributing to a comprehensive understanding of its operational characteristics.

In conjunction with the incubator parameters, the weight and gestational age of the infant are utilized to forecast the infant's breath rate and breath volume. This prediction is informed by reference to analogous cases involving infants exhibiting comparable developmental milestones, thereby enhancing the prognostic capabilities of the model.

4.2 The prediction of breath rate and breath volume is based on:

4.2.1 Age of the infant: The gestational period in weeks is the age of the infant , greater the gestational age ,greater is the health of the infant ,therefore the breath rate would be more sustained .With the predicted value of breath rate , in case of any variation in breath rate for the child, serious ailments can be identified at an earlier stage.

4.2.2.weight of the infant: The weight in grams of the infant is also a rough estimate of its overall growth and development, which can be used to estimate the breath rate and the breath volume.

As mentioned previously, the humidity inside the incubator is measured using the fabricated sensor and the temperature using dht11, therefore the parameters currently present inside the incubator are measured and displayed and the predicted values are also displayed, and immediately stored to the cloud. The system is also connected to a telegram bot ,so even when the medicinal practitioner is at a far off location, they would have access to the data through

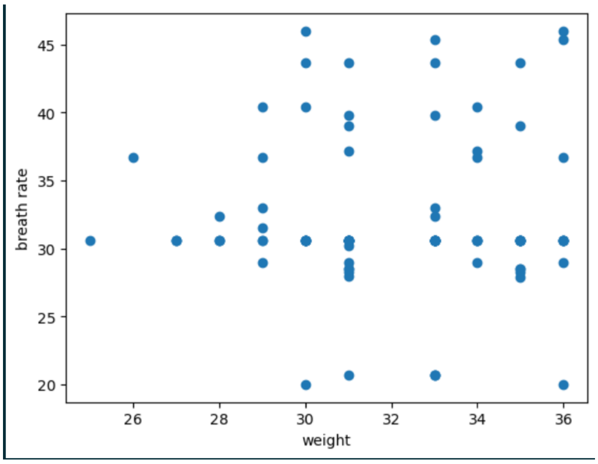


Fig 4.2.1 Relation between weight and breath rate

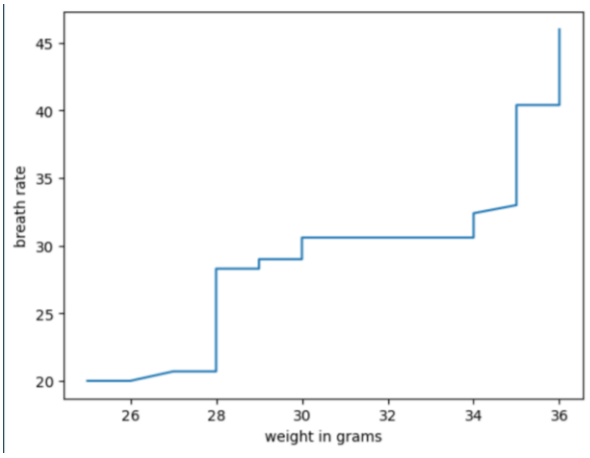


Fig 4.2.1 Relation between weight and breath rate

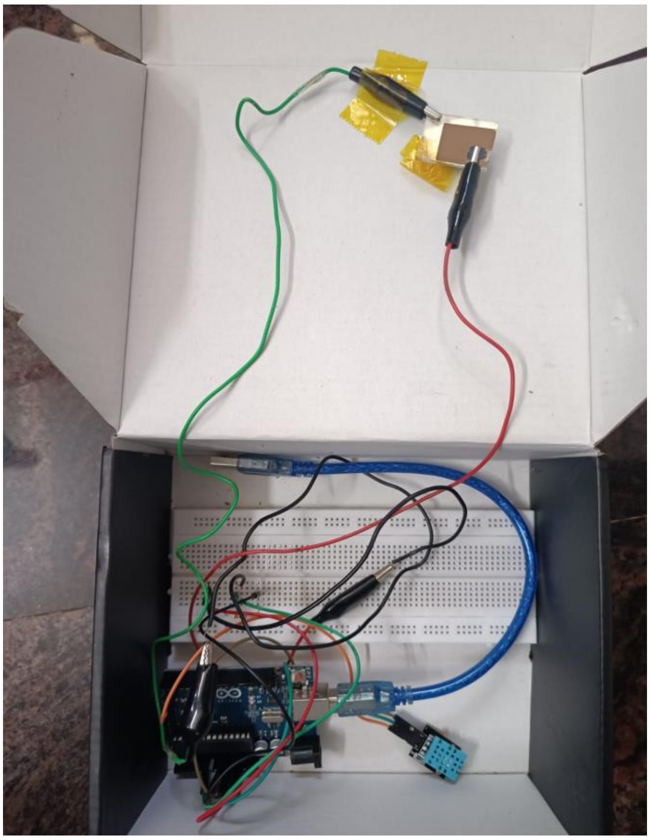
the telegram bot or through the cloud directly.

The incubator parameters which should be set for a particular child by taking the weight,age,room temperature,room humidity .

Model Evaluation:

Regression Model	Value measured									
	Breath rate					Breath volume				
	MSE	RMSE	MAE	R^2	Acc.	MSE	RMSE	MAE	R^2	Acc.
Linear Reg.	30.3367	5.50788	3.89	0.73	0.94	48.6866	6.977	48.42	0.92	0.80

4.2:The external circuit connected:



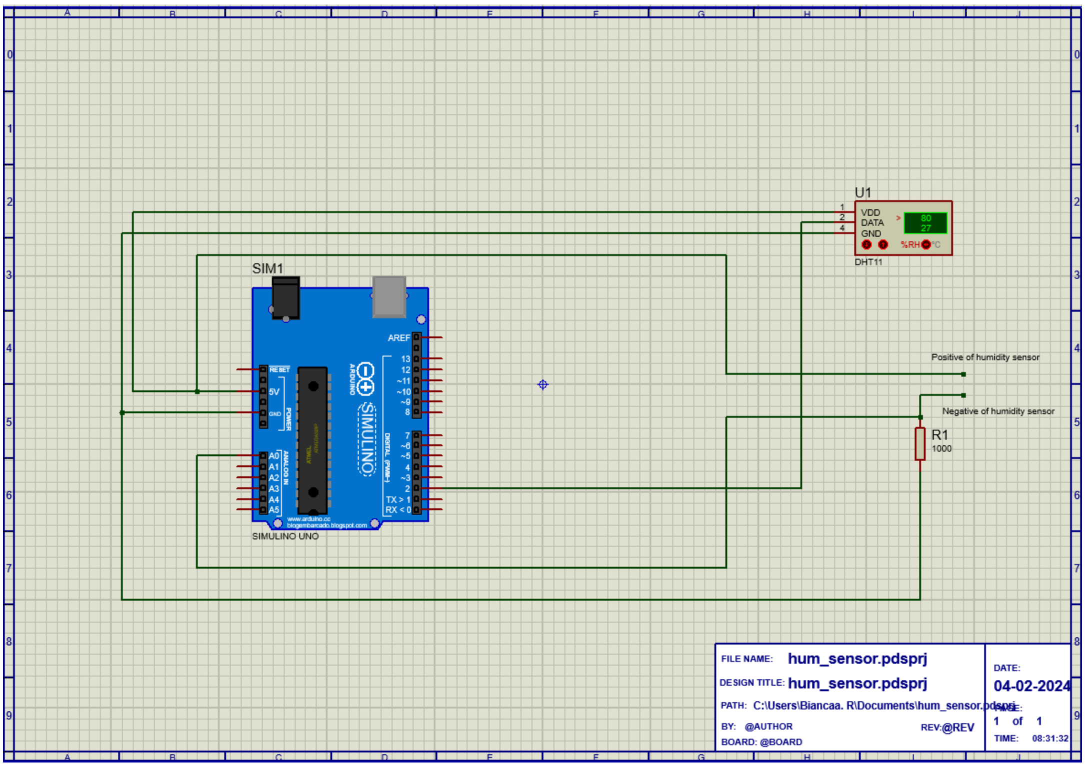
The schematic representation of the external circuit is depicted, wherein the Arduino unit is energized through connection to a laptop. The DHT11 sensor is integrated and energized via the 3.3V power source. A stable 5V is supplied as input to the reference resistance and the fabricated humidity sensor combination. The analog pin A0 is designated for reading the voltage differential across the fabricated humidity sensor. The resultant value is within the range of 0 to 1023 and is subsequently converted from a 5V scale.

The preference for the use of Arduino instead of other edge devices would be that, since it’s a cost effective microcontroller it is more suitable for this purpose and employing a microprocessor like raspberry pi would be unnecessary for the tiniest of computational tasks.

Fig 4.2.1 External circuit

The module PyQt5 used for GUI is more suitable to be worked with Arduino as it could continue running in the host system, on the other hand if it was enabled in a raspberry pi it would have been run on Debian os which has certain GUI functionalities different. The analog to digital pin available in Arduino was another reason to gravitate towards it as there would not be a need to establish an external analog to digital converting circuit.\

Predicated upon the discerned voltage drop across the sensor and the drop across the established reference resistance, the resistance of the humidity sensor can be approximated. This derived resistance value is then input into the machine learning model to facilitate predictions pertaining to the measured relative humidity. This comprehensive circuit configuration enables the acquisition of precise sensor data, subsequently utilized for predictive modeling of environmental humidity levels.



4.2.2 Circuit diagram in proteus

4.2.2 Calculation involved:

sensorValue = analogRead(sensorPin); // Read Vout on analog input pin A0 (Arduino can sense from 0-1023, 1023 is 5V)

Vout = (Vin * sensorValue) / 1023; // Convert Vout to volts

R= (((Vin* Rref)/Vout)-Rref);

Where V ref is 1000 ohms (Taken in accordance to the dataset of the sensor)

Therfore by multiplying the analog output value's fraction with 5v a voltage drop in the range
(0-5 v) is found, Using which the current value of attained resistance is obtained.

4.3 Application Development:

The developed application is a desktop application created entirely using the Python programming language, with the integration of the PyQt5 module for the graphical user interface (GUI) and MySQL for database interaction.

4.3.1 Features in the Application:

1. **Record Register:** The application incorporates a record register, which archives and retrieves all historical data from the cloud. This feature ensures accessibility to previously added and stored data through the application.
2. **Authentication System:** A secure signup and login mechanism is implemented, integrating with a database to restrict unauthorized access and safeguard sensitive information.
3. **Real-time Prediction:** The application provides a functionality to add new records and receive immediate prediction results. The machine learning model operates in the backend, contributing to real-time data analysis.
4. **Customized Dashboard:** Each user is endowed with a personalized dashboard tailored to individual preferences and requirements, enhancing user experience and facilitating efficient data management.
5. **Telegram Bot Integration:** The application seamlessly integrates with a Telegram bot, allowing convenient data access. Medical professionals, such as doctors, can effortlessly retrieve data via the Telegram bot by submitting requests.
6. **Cloud Integration (ThingSpeak):** The application ensures seamless and immediate transmission of added values to the cloud. The chosen cloud service for this purpose is ThingSpeak, facilitating robust and glitch-free data storage and retrieval. This cloud integration enhances data accessibility and facilitates remote monitoring.

4.3.2: The user interface:

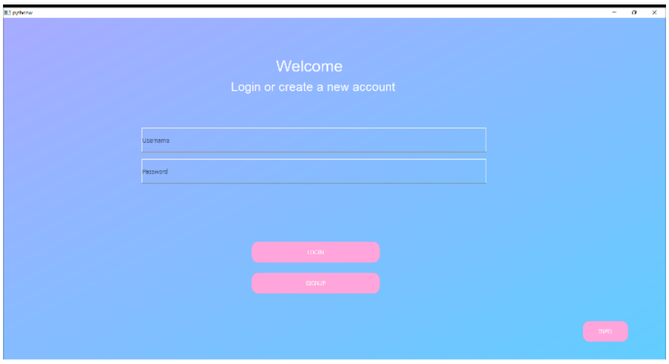


Fig 4.3.2.1 login screen

Fig 4.3.2.2 signup screen

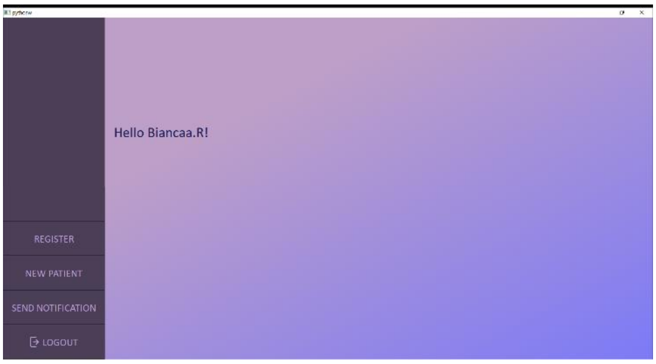


Fig 4.3.2.3 customized dashboard

Fig 4.3.2.4 Older records

Fig 4.3.2.5 Adding new inputs

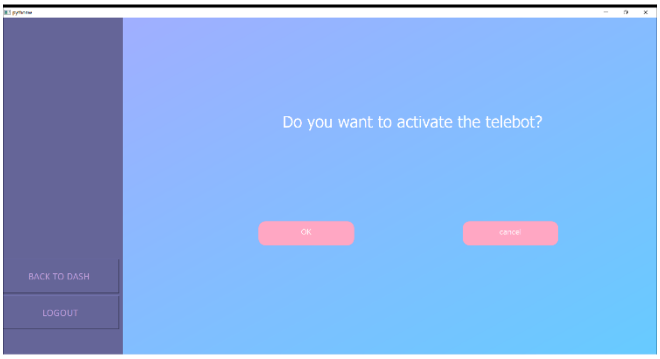
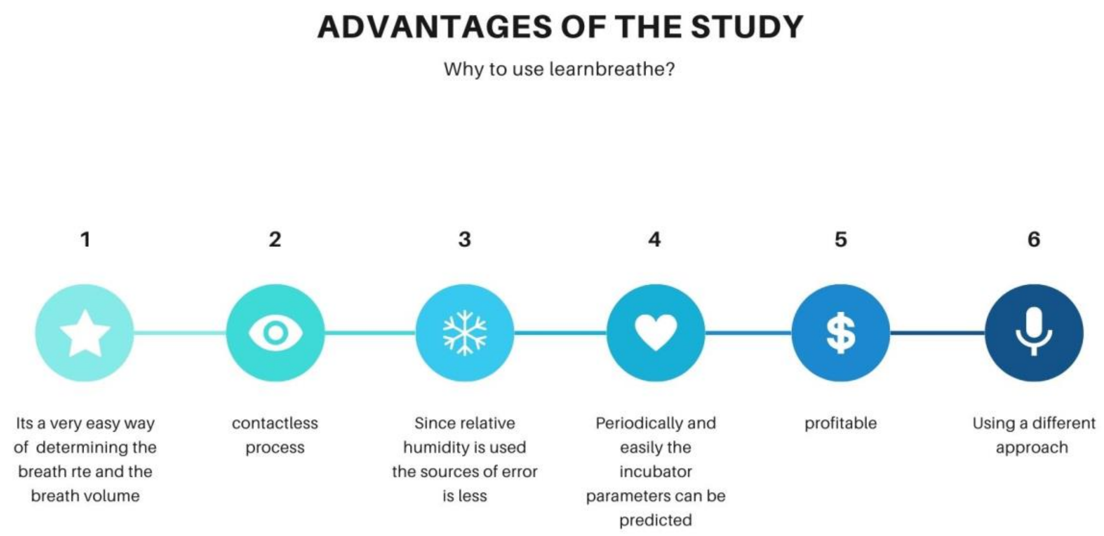


Fig 4.3.2.6 Telepage for activating the telegrambot

Fig4.3.2.7 Prototype of the proposed system

5:APPLICATION:

The application of this study is that is can be used in hospitals and if used in a largescale, a great number of premature babies, who die each year can be saved.



6.CONCLUSION:

In conclusion, this study was undertaken to examine the conditions surrounding preterm infants and propose strategies to enhance their longevity by predicting optimal incubator settings, specifically temperature and humidity, to maximize their likelihood of survival. The prediction model incorporates breath rate and breath volume data from infants with similar developmental profiles, facilitating the identification and analysis of any deviations in the respiratory patterns of a given child. The overarching objective of this study is to mitigate the incidence of mortality attributed to preterm births.

7.REFERENCES:

1. Dive, K. and Kulkarni, G. (2013). Design of Embedded Device for Incubator for the Monitoring of Infants. Int. Journal of Advanced Research in Computer Science and Software Engineering, Vol. 3, Issue 1, pp. 541-546

2. Arduino Based Baby Incubator Using GSM Technology
Atul W. Kale1,Ajay H. Raghuvanshi2, Pranit S. Narule3, Pooja S. Gawatre4 , Shilpa Surwade5
3.Boutourline-Young, H. J. & Smith, C. A. (1950). Amer. J. Di&. Child. 80,753. Cross, K.W. (1949). J. Phy8iol. 109, 459. Cross, K. W. & Warner, Pamela (1951). J. Phy8iol. 114, 283. Roberts, P. W. (1950). J. mci. Inatrum. 27, 105. Wilson, J. L., Reardon, Helen S. & Murayama, M. (1948). Paediatric8, 1, 581.

4. An Android Application Based Temperature and Humidity Monitoring and Controlling System for Child Incubators article in International Journal of Scientific and Engineering Research · January 2018

[5] R. Cichocki, R. O'Laughlin, A. Midouin, B. F. Busha, DzTransportable)nfant)ncubato For Developing Countriesdz, IEEE 978-1-4693-1142,Dec 2012.

[6´] N.S. Joshi, R.K. Kamat, and P.K. Gaikwad, DzDevelopment of Wireless Monitoring System for Neonatal Intensive Care Unitdz,)JACR, Volume-3 Number-3 Issue-11 September-2013.

[7] R. Ramanig, S. Valarmathy, S. Selvaraju and P. Niranjan, DzGSM Technologydz.)nternational Journal of Computer Application (0975-8887), Volume 57-No.18, November 2012.
[8] Med A.z., Elyes f. , and Abdekader m., DzAmerican journal of Engineering and Applied Science Dz, Application of Adaptive Predictive Control to a Newborn Incubator, 4 (2) : 235-243,
ISSN 1941-7020, 2011
[9] M. Suruthi, S.Suma.dzMicrocontroller Based Baby Incubator Using Sensordz,îAn)SO ˛´97:´TT7 Certified Organization) Vol.4,Issue 12, December 2015.

[10] Prof. Kranti Dive*, Prof. Gitanjal8i Kulkarni Computer Engineering Dept. M)T College of Engineering Pune,)ndia Dz Design of Embedded Device for Incubator for the Monitoring of)nfantsdz,)JARCSSE All Rights Reserved Page | иИТvolume
3, Issue 11, November 2013.

[11} Non-contact physiological monitoring of preterm infants in the Neonatal Intensive Care Unit Mauricio Villarroel1*, Sitthichok Chaichulee 1 , João Jorge 1 , Sara Davis2 , Gabrielle Green2 , Carlos Arteta3 , Andrew Zisserman 3 , Kenny McCormick2 , Peter Watkinson4 and Lionel Tarassenko

[12] [12]Multiple Fault Detection and Smart Monitoring System Based on Machine Learning Classifiers for Infant Incubators Using Raspberry Pi 4(using a microprocessor)
Maryam A. Mahdi*, Sabah A. Gittaffa, Abbas H. Issa
[13] Shabeeb, A.G., Al-Askery, A.J., Nahi, Z.M. (2020). Remote monitoring of a premature infants incubator. Indonesian Journal of Electrical Engineering and Computer Science, 17(3): 1232–1238. https://doi.org/10.11591/ijeecs.v17.i3

BUDGET SPLITUP:

I. DEVICE FABRICATION METHODOLOGIES

ZnO/NiO nanomaterial synthesis

As shown in Fig. 2, 80 ml of deionized water is first mixed with 0.5 mmol of zinc acetate dihydrate and 0.25 mmol of nickel acetate tetrahydrate, as well as 0.276 g of SDS and 0.437 g of CTAB. The resulting mixture is then heated for 6 hours at 130°C in a Teflon-lined stainless steel autoclave. The precursor is then calcined for 4 hours at 350°C using a furnace. After cooling naturally, a black color ZnO-NiO composite powder is collected.

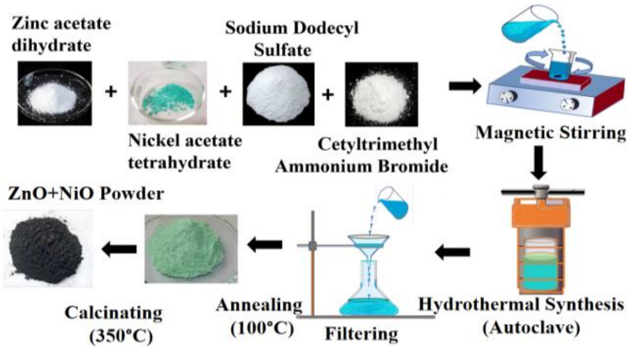


Fig. 2. Hydrothermal Synthesis of SDS:CTAB mixed ZnO/NiO Nanomaterial

Fabrication of the Sensitive Thin Film

The glasses coated with fluorine-doped tin oxide (FTO) are put to use for the thin-film deposition process. The binders used to crush and make the calcined composite powder are ethylcellulose and alpha tripinol. The solution is spin-coated for 10 seconds at 3000 rpm after 10 hours to create a thin sensing layer.

II. MATERIAL CHARACTERIZATION TECHNIQUES

A. X-Ray Diffraction (XRD)

Utilizing XRD, the structure and phase purity of synthesized nanoparticles are evaluated.

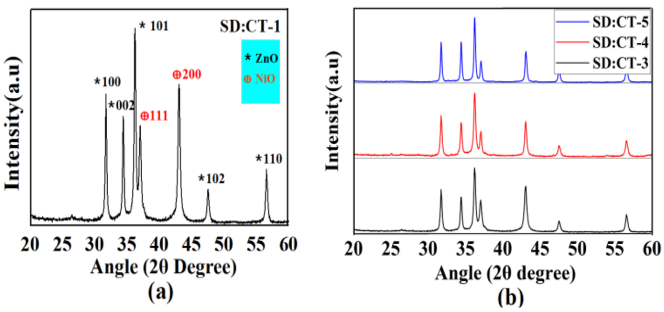


Fig. 3. XRD Pattern of (a) Intensity peaks of SD:CT-1, (b) Integrated pattern Samples

B. Morphological Characterization

FESEM is used to examine the size and morphology of the samples. In Fig. 4, FESEM images of ZnO/NiO at different surfactant molar ratios are displayed.

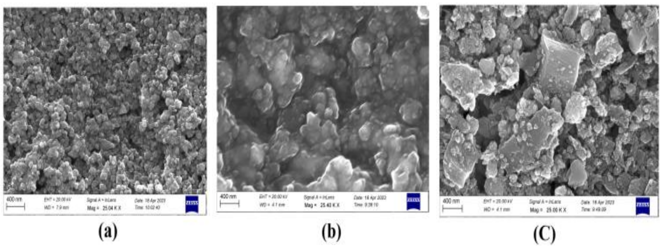


Fig. 4. FESEM image of (a) SD:CT-3 device (b) SD:CT-4 device (c) SD:CT-5 device at 400 nm, 25 K magnification

C. Brunauer- Emmett-Teller (BET) Surface area analysis

The specific surface area (SBET), pore volume, and average diameter Barret-Joyner-Halenda (BJH) of the various concentrations of surfactant mixed ZnO/NiO samples are calculated using nitrogen (N_2) adsorption. The BET equation is given by

$$\frac{1}{W((\frac{P_o}{P})-1)} = \frac{1}{W_m C} + \frac{C-1}{W_m C}\left(\frac{P}{P_o}\right) \qquad (1)$$

Where, W is the weight of adsorbed gas, $\frac{P}{P_o}$ is the relative pressure, W_m is the weight of adsorbate as a monolayer, and C is the BET surface area constant

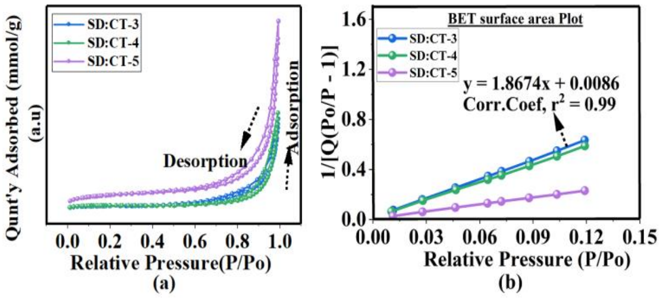


Fig. 5. (a) Nitrogen adsorption-desorption isotherms,
(b) BET surface area plot,

D. Current - Voltage characteristics

In the accompanying Fig. 6 (a), (b), and (c), the p-n junction rectifying characteristics of a surfactant-based ZnO/NiO nanosensor are shown, respectively, in the dark and direct sunlight

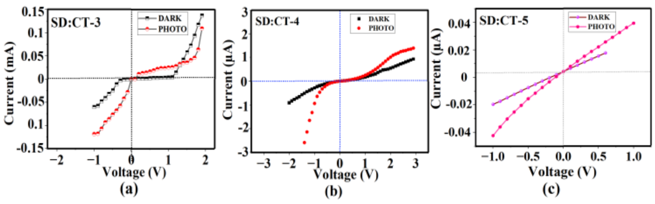


Fig. 6 I-V characteristics of the proposed devices in the dark and photo condition (a) SD:CT-3 device (b) SD:CT-4 device (c) SD:CT-5 device

BILL SPLITUP:

S:No.	Description	Qn'ty	Ql'ty	Invoice No. & Date	Receiving Date	Make	Remarks
1	Indicator Papers-Full Range (PH 1.0 to 14.0) 10 BKS Fisher scientific, EthanolLarChinaSqlabuse	2	OK	VS/C2021/23-24	27.05.23	---	NIL

2	Sathyabama Institute of Science and Technology	7	OK	2456	12.07.23	---	NIL

S:No.	Description	Qn’ty	Ql’ty	Invoice No. & Date	Receiving Date	Make	Remarks
1	Main components like Arduino, raspberry pi ,bread board	1	OK	Invoice No.: 4537 Date: 29-05-2024	Invoice No.: 4537 Date: 29-05-2024	---	NIL

IFSP

student :- BIANCAA

Sri Sivasubramaniya Nadar College of engineering

Claim reimbursement/Release of amount under Sponsored (Int/~~Ext~~) Projects

Submission date: 25.5.24 Sanctioned amount:20,000

ERP ID	2210329
Name of the PI	Dr. B.Partibane
Duration of the project	Twelve months
Mail ID and mobile	partibaneb@ssn.edu.in
Department	ECE
Sponsored project*	Analytical approach of the effect of surfactant on mixed metal oxide materials (Zno/Nio) to enhance the sensitivity of a humidity sensor
Claim under category	Materials
Remarks -for overhead charges(purpose)	Buying of components

*Strike out which is not applicable

Project reference number	Balance eligible amount	Total amount claimed so far	Original bill number and date	Amount requested for reimbursement/release to the vendor	Remaining balance
3	3678	16322	BY Electronics Invoice No.: 4537 Date: 29-05-2024	3678	None

Amount spent according to bill: 5575 rs

*Since the amount crosses the budget, the amount to be claimed : 3678 only

B.Partibane	02/07/24	P.Ramesh
Signature of the PI	Approval by HoD	Dean research (IFSP)

*IFSP Only Dean(Research)Signature is required

Vice principal	Manager(Procurement)

Sanctioned Internally Funded Projects for First Year UG/PG Students 2023

SRI SIVASUBRAMANIYA NADAR COLLEGE OF ENGINEERING KALAVAKKAM-603110

ECE - UG STUDENT PROJECTS

S. No	Name and Year of the Student(s)	Project Guide(s)	Title of the Project	Duration	Budget & Items Approved
1	Aathi Shankar K.S. Jayanth J. Sanjay Kumar	Dr. M. Srinivasan SSN RC	Enhancing the thermoelectric properties if mineral acanthite (Ag_2S) by filler compounds for wearable thermoelectric generators	2 Years	**Budget: Rs.25000** **Items Approved:** 1. Chemicals 2. Characterization 3. Miscellaneous
2	I. Gowshika R. Prithika	Dr. P. Rajesh Physics	Cobalt-Manganese-Zinc Oxide on silicon carbide composite electrode for fabrication of symmetric supercapattery	1 Year	**Budget: Rs.20000** **Items Approved:** 1. Chemicals 2. Miscellaneous
3	R. Biancaa	~~Dr. S. Kirubaveni~~ Dr. B. PARTIBAN	Analytical approach of the effect of surfactant on mixed metal oxide materials (ZnO/NiO) to enhance the sensitivity of a humidity sensor	1 Year	**Budget: Rs.20000** **Items Approved:** 1. BET Studies 2. FESEM and XPS Studies 3. Chemicals 4. Miscellaneous

SSN

BY Electronics

No-5, Gandhi Road, West Tambaram, Chennai
Phone no.: 7904425459
Email: byelectronicschennai@gmail.com
GSTIN: 33BBZPN3771L1ZU
State: 33-Tamil Nadu

Tax Invoice

Bill To

Biancaa.R

SSN College of Engineering

Invoice Details

Invoice No.: 4537
Date: 29-05-2024

#	Item name	HSN/ SAC	Quantity	Price/ unit	GST	Amount
1	**Arduino UNO R3**	84733020	1	₹ 750.00	₹ 135.00 (18.0%)	₹ 885.00
2	**DHT 11 Sensor Module**	90312000	1	₹ 100.00	₹ 18.00 (18.0%)	₹ 118.00
3	**1/4W Resistor box**	85366990	1	₹ 40.00	₹ 7.20 (18.0%)	₹ 47.20
4	**800 points Breadboard**	8536	1	₹ 100.00	₹ 18.00 (18.0%)	₹ 118.00
5	**Jumper Wires Set (M-M, F-F, F-M)**	85441990	2	₹ 40.00	₹ 14.40 (18.0%)	₹ 94.40
6	**9V Battery (HW)**	8506	1	₹ 40.00	₹ 7.20 (18.0%)	₹ 47.20
7	**9V Battery cap**	8538	1	₹ 10.00	₹ 1.80 (18.0%)	₹ 11.80
8	**BC547 transistor**	85412100	5	₹ 5.00	₹ 4.50 (18.0%)	₹ 29.50
9	**1N4007 diode**	8541	5	₹ 2.00	₹ 1.80 (18.0%)	₹ 11.80
10	**SG90 Servo Motor**	85011019	1	₹ 180.00	₹ 32.40 (18.0%)	₹ 212.40
11	**LM317 IC**	8542	1	₹ 60.00	₹ 10.80 (18.0%)	₹ 70.80
12	**1602 LCD Display Green**	85423100	1	₹ 200.00	₹ 36.00 (18.0%)	₹ 236.00
13	**Raspberry Pi ZERO W**	8471	1	₹ 2,820.00	₹ 507.60 (18.0%)	₹ 3,327.60
14	**LED (red, green) 5mm**	8541	10	₹ 2.00	₹ 3.60 (18.0%)	₹ 23.60
15	**L293D Motor driver**	85011019	1	₹ 290.00	₹ 52.20 (18.0%)	₹ 342.20
	Total		**33**		**₹ 850.50**	**₹ 5,575.50**

Invoice Amount In Words

Five Thousand Five Hundred and Seventy Five Rupees and Fifty Paisa only

Terms And Conditions

Thank you for doing business with us.

Sub Total	₹ 4,725.00
SGST@9.0%	₹ 425.25
CGST@9.0%	₹ 425.25
Total	**₹ 5,575.50**
Received	₹ 5,575.50
Balance	₹ 0.00

Pay To:

Bank Name: ICICI BANK LIMITED
Bank Account No.: 139305500730
Bank IFSC code: ICIC0001393
Account Holder's Name: BY Electronics

For: BY Electronics

Authorized Signatory

BY Electronics

No-5, Gandhi Road, West Tambaram, Chennai
Phone no.: 7904425459
Email: byelectronicschennai@gmail.com
GSTIN: 33BBZPN3771L1ZU
State: 33-Tamil Nadu

BY Electronics

Tax Invoice

Bill To

Biancaa.R

SSN College of Engineering

Invoice Details

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Date: 29-05-2024

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9	1N4007 diode	8541	5	₹ 2.00	₹ 1.80 (18.0%)	₹ 11.80
10	SG90 Servo Motor	85011019	1	₹ 180.00	₹ 32.40 (18.0%)	₹ 212.40
11	LM317 IC	8542	1	₹ 60.00	₹ 10.80 (18.0%)	₹ 70.80
12	1602 LCD Display Green	85423100	1	₹ 200.00	₹ 36.00 (18.0%)	₹ 236.00
13	Raspberry Pi ZERO W	8471	1	₹ 2,820.00	₹ 507.60 (18.0%)	₹ 3,327.60
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15	L293D Motor driver	85011019	1	₹ 290.00	₹ 52.20 (18.0%)	₹ 342.20
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Balance	₹ 0.00

Pay To:

Bank Name: ICICI BANK LIMITED
Bank Account No.: 139305500730
Bank IFSC code: ICIC0001393
Account Holder's Name: BY Electronics

For: BY Electronics

Authorized Signatory

Date:03.08.23
Kalavakkam

From,
Dr.B.Partibane
Associate Professor
Department of ECE
SSN College of Engineering
Kalavakkam-603110

To
The Vice Principal
SSN College of Engineering
Kalavakkam-603110

Respected madam,

Sub: Reimbursement of amount spent towards internal student funding project - Reg.,

Project Title Application:Analytical approach of the effect of surfactant on Mixed metal Oxide Materials(ZnO/NiO) to enhance the sensitivity of a humidity sensor

Name of the Student: Biancaa. R: Biancaa.R

Budget: 20000/-

This letter is to seek your approval for the reimbursement of Rs.16322/- that has been spent towards the purchase of Indicator Papers-Full Range (PH 1.0 to 14.0) 10 BKS Fisher scientific, EthanolLarChinaSqlabuse (500 ml)-9802, and FESEM/EDAX Tests. The original bills and selection letter has been enclosed.

S:NO	COMPANY NAME	BILL NO.	BILL DATE	AMOUNT
1	Vijaya Scientific Company	VS/C2021/23-24	27.05.23	922
2	Sathyabama Institute of Science and Technology	2456	12.07.23	15400
			Total	**16322**

Thanking you,

Forwarded
Vijayalakshmi
03/08/23
P.Ramalay

Yours sincerely

Dr.B.Partibane

7/8/23

SSN College of Engineering
Kalavakkam-603110

Quality and Quantity Inspection Report for Items received

P.O.No.& Date : NIL

Suppliers Name : Vijaya Scientific Company, Sathyabama Institute of Science & Technology

S:No .	Description	Qn'ty	Ql'ty	Invoice No. & Date	Receiving Date	Make	Remarks
1	Indicator Papers-Full Range (PH 1.0 to 14.0) 10 BKS Fisher scientific, EthanolLarChinaSqlabuse	2	OK	VS/C2 021/23 -24	27.05.23	---	NIL
2	Sathyabama Institute of Science and Technology	7	OK	2456	12.07.23	---	NIL

Signature of the Project Guide

Signature of The HOD

SSN College of Engineering
Kalavakkam-603110

Quality and Quantity Inspection Report for Items received

P.O.No.& Date : NIL

Suppliers Name : Vijaya Scientific Company, Sathyabama Institute of Science & Technology

S:No .	Description	Qn'ty	Ql'ty	Invoice No. & Date	Receiving Date	Make	Remarks
1	Indicator Papers-Full Range (PH 1.0 to 14.0) 10 BKS Fisher scientific, EthanolLarChinaSqlabuse	2	OK	VS/C2 021/23 -24	27.05.23	---	NIL
2	Sathyabama Institute of Science and Technology	7	OK	2456	12.07.23	---	NIL

Signature of the Project Guide

Signature of The HOD

Sanctioned Internally Funded Projects for First Year UG/PG Students 2023

SRI SIVASUBRAMANIYA NADAR COLLEGE OF ENGINEERING
KALAVAKKAM-603110

ECE - UG STUDENT PROJECTS

S. No	Name and Year of the Student(s)	Project Guide(s)	Title of the Project	Duration	Budget & Items Approved
1	Aathi Shankar K.S. Jayanth J. Sanjay Kumar	Dr. M. Srinivasan SSN RC	Enhancing the thermoelectric properties if mineral acanthite (Ag_2S) by filler compounds for wearable thermoelectric generators	2 Years	**Budget: Rs.25000** **Items Approved:** 1. Chemicals 2. Characterization 3. Miscellaneous
2	I. Gowshika R. Prithika	Dr. P. Rajesh Physics	Cobalt-Manganese-Zinc Oxide on silicon carbide composite electrode for fabrication of symmetric supercapattery	1 Year	**Budget: Rs.20000** **Items Approved:** 1. Chemicals 2. Miscellaneous
3	R. Biancaa	Dr. S. Kirubaveni	Analytical approach of the effect of surfactant on mixed metal oxide materials (ZnO/NiO) to enhance the sensitivity of a humidity sensor	1 Year	**Budget: Rs.20000** **Items Approved:** 1. BET Studies 2. FESEM and XPS Studies 3. Chemicals 4. Miscellaneous

Personnel Copy

SATHYABAMA

INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)
Accredited with 'A' grade by NAAC
Jeppiaar Nagar, Rajiv Gandhi Salai, Chennai - 600 119.

DNV GL
ISO 9001:2015

Centre for Nanoscience & Nanotechnology

Consultancy fee Receipt

No. 2456 Date: 12/7/23

We received with thanks from ....The Principal........

of........SSN College of Engg........

a sum of Rs....15400.....(In Words ....fifteen thousand

four hundred..........only) towards analyzing the samples XRD/

EDAX/FESEM/EDAX/HRSTEM/SAED/AFM/RAMAN/Epi.Mic./SONICATOR

/RF/ DC Sputtering/Evaporation/ UV Thickness/ Coatings / Thickness/ Stress/

Roughness/ Polishing/ Cutting/ Ball Mill/ Optical Microscope /Tribo Meter/

/Corrosion/FTIR/UV-Vis/Hall Effect at Centre for Nano science and Technology,

Sathyabama Institute of Science and Technology, Chennai -119.

No. of Samples: 7

Rs.....15400/-

Dr.T.SASIPRABA, M.E.,Ph.D.,
VICE CHANCELLOR
SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)
Jeppiaar Nagar, Rajiv Gandhi Salai,
Chennai-600 119.

Authorized Signatory

CamScanner

TAX INVOICE *(ORIGINAL FOR RECIPIENT)* e-Invoice

IRN : 3a1e427a7d9e7023ce776f31d038bee1f6c5bc88a6ea78a-8a613f6d8d6e25973
Ack No. : 152314733677717
Ack Date : 27-May-23

Vijaya Scientific Company
Head Office -
Old No: 1/422, New No: 1/597
Vinayagar Nagar, 5th Street,
Thuraipakkam, Chennai - 600 097
Ph: 044- 24560521, 24561639
Godown -
No 1/1, T.P.G Illam, Rajiv Gandhi Street
Thazambur Thirupporur Taluk
Kanchipuram, Tamilnadu - 600 130
GSTIN/UIN: 33ACPPL8181L1ZZ
State Name : Tamil Nadu, Code : 33
E-Mail : vijayascientific08@gmail.com

Invoice No. VS/C2021/23-24	Dated 27-May-23
Delivery Note	Mode/Terms of Payment
Reference No. & Date.	Other References Poundoss
Buyer's Order No.	Dated
Dispatch Doc No.	Delivery Note Date
Dispatched through	Destination
Terms of Delivery	

Buyer (Bill to)
The Principal,
S.S.N College of Engineering,
Rajiv Gandhi Salai, (OMR)
Kalavakkam-603 110
GSTIN/UIN : 33AAATS2240Q1ZD
State Name : Tamil Nadu, Code : 33

Sl No.	Description of Goods	HSN/SAC	GST Rate	Quantity	Rate	per	Disc. %	Amount
1	**38141 '10140' Indicator Papers - Full Range (PH 1.0 to 14.0) 10 BKS Fisher Scientific**	38229090	12 %	**2.00 Nos**	280.00	Nos		560.00
2	**ETHANOLARCHINASQLABUSE 500ML -9802**	98020000	18 %	**1.00 Nos**	250.00	Nos		250.00
								810.00
	Output CGST - 6%				6	%		33.60
	Output SGST - 6%				6	%		33.60
	Output CGST - 9%				9	%		22.50
	Output SGST - 9%				9	%		22.50
	Rounded Off							(-)0.20
	Less :							
	Total			**3.00 Nos**				**₹ 922.00**

E. & O.E

Amount Chargeable (in words)
INR Nine Hundred Twenty Two Only

Company's Bank Details
Bank Name : **IDFC Bank - 10068582076**
A/c No. : **10068582076**
Branch & IFS Code : **EGATTUR & IDFB0080107**

Company's PAN : **ACPPL8181L**

Declaration
We declare that this invoice shows the actual price of the goods described and that all particulars are true and correct.

for Vijaya Scientific Company

VIJAYA SCIENTIFIC COMPANY CHENNAI 600 097

Authorised Signatory

This is a Computer Generated Invoice

S.R.

29/5/23

Secy.

SEC[illegible]

Sr No 14557/[illegible]

Vehicle No TN 14 BA 9330

Date 29/05/[illegible]

Time - In / Out ___ a.m. / p.m.